Bulletin no. / Nr.	Issue date / Ausgabedatum	Revision
SB-2025-009	**2025-06-18**	**0**
Product / Produkt	Supersedes / Ersetzt	Compliance / Umsetzung
HCP-350 centrifugal pump	**-**	**Mandatory (see 5)**

Kategorie / Category: [X] **Safety** [ ] Reliability [ ] Information

HCP-350 from serial HCP350-240001: revised casing cover and casing gasket. New casing bolt tightening torque and inspection of pumps in service.

HCP-350 ab Seriennummer HCP350-240001: geänderter Gehäusedeckel und Gehäusedichtung. Neues Anzugsmoment der Gehäuseschrauben und Prüfung der Pumpen im Betrieb.

1 Summary / Zusammenfassung

Since production week 01/2024 (serial number HCP350-240001), HCP-350 pumps are built with a revised casing cover and a thinner casing gasket. The tightening torque for the casing bolts on these pumps is 38 N-m, applied in a cross pattern in two passes. The value of 45 N-m given in the Installation, Operation and Maintenance Manual HDN-IOM-350 Rev D applies only to pumps with serial numbers before HCP350-240001.

Tightening a revised casing cover to 45 N-m overloads the gasket seat and can crack it. A cracked gasket seat cannot be repaired. Pumps in service whose casing joint has been opened since delivery must be inspected as described in 5.2.

2 Affected products / Betroffene Produkte

Model	Serial number range	Casing cover	Casing gasket	Casing bolt torque
HCP-350	HCP350-210001 to HCP350-239999	Version 1 (original)	1.5 mm fibre	45 N-m, per IOM Rev D (unchanged)
HCP-350	HCP350-240001 and later	Version 2 (revised)	0.8 mm aramid fibre, graphite faced	38 N-m, per this bulletin
HCP-200	all	-	-	Not affected

Identification. Version 2 casing covers carry the cast mark "V2" on the flange next to the drain plug boss, and the raised gasket face is 8 mm wide instead of 12 mm. The nameplate does not show the casing version. If the nameplate is missing or not readable, use the cast mark.

Replacement casing covers supplied as spare parts after January 2024 are always Version 2, whatever the serial number of the pump. A pump before HCP350-240001 that has been fitted with a replacement cover is to be treated as a Version 2 pump for all purposes of this bulletin. Record the cover change on the pump's maintenance record.

3 Symptom / Schadensbild

Since October 2024 Halden has received 14 field reports from North America and Europe of weeping at the joint between casing and casing cover on HCP-350 pumps less than two years old. In all reported cases the leakage started within days or weeks after the casing joint had been opened in the field, most often for replacement of the mechanical seal (IOM 8.4), in two cases for wear ring replacement.

On the returned covers, dye penetrant testing showed radial hairline cracks in the gasket seat, starting at one or more bolt holes and running toward the bore. Leakage is small at first (drops, glycol residue below the joint) and increases with thermal cycling. No case has been reported on a pump whose casing joint had not been opened since delivery.

4 Root cause / Ursache

The Version 2 casing cover was introduced to reduce casting weight and to improve casting yield at the foundry. The gasket seat was narrowed and the casing gasket changed from 1.5 mm fibre to 0.8 mm aramid fibre with graphite facing. Pumps are assembled in the factory with 38 N-m. The change was released in production without a corresponding revision of the operating instructions, so the IOM still states 45 N-m in sections 4.3, 8.4 and 8.7.

With the thinner gasket and the narrower seat, a bolt torque of 45 N-m produces a surface pressure on the seat above the permissible value for the casting. The overload alone does not always cause immediate leakage. Cracks initiate at the bolt holes and grow with each heating and cooling cycle. Use of impact wrenches or tightening without a cross pattern increases the risk further.

Halden regrets that the documentation did not follow the production change. The IOM will be corrected in Rev E (see 7).

5 Corrective action / Massnahmen

5.1 Every time the casing joint of an affected pump is closed

1. Read the pump serial number and check the cast mark on the cover (see 2). Treat any Version 2 cover as affected.
2. Clean the gasket seat on casing and cover. With the gasket removed, inspect the seat for cracks, preferably with dye penetrant. A cracked cover must be replaced (5.2, step 5).
3. Fit a new Version 2 casing gasket. Never reuse a gasket. Never fit a 1.5 mm Version 1 gasket on a Version 2 seat, or the reverse; the joint will not seal.
4. Lightly oil the bolt threads (property class 8.8). Do not use anti-seize paste with a lower friction value; it raises the clamping force at the same torque.
5. Tighten the casing bolts in a cross pattern in two passes: first pass to approximately 20 N-m, second pass to 38 N-m. Use a calibrated torque wrench. Impact tools are not permitted on this joint.
6. After the first 24 hours at operating temperature, check the torque once at 38 N-m. Do not re-tighten beyond this value to stop a leak; a joint that leaks at 38 N-m has a damaged gasket or seat.
7. Record the torque applied and the reference "SB-2025-009" in the maintenance record (IOM Annex B).

5.2 Field inspection of affected pumps in service

Carry out this inspection at the next scheduled maintenance visit, at the latest by 2025-12-31, on all pumps from serial HCP350-240001 (and on earlier pumps fitted with a Version 2 cover) whose casing joint has been opened since delivery. Pumps whose casing joint has never been opened were assembled at the correct torque and need no action.

1. Check the maintenance records for the pump: was the casing joint opened, and which torque was applied? If the records do not state the torque, assume 45 N-m.
2. With the pump running at operating pressure and temperature, inspect the casing joint all round for weeping, glycol or mineral residue, and staining on the casing below the joint.
3. No leakage and 38 N-m documented: no further action. Record the inspection.
4. No leakage but 45 N-m documented or torque unknown: at a convenient stop, isolate and drain the pump (IOM 2.4), slacken the casing bolts, and re-tighten to 38 N-m per 5.1, step 5. Inspect again after 24 hours at operating temperature. If leakage appears, continue with step 5.
5. Leakage present: stop, isolate and drain the pump. Open the casing joint, remove the gasket and inspect the gasket seat with dye penetrant. Without cracks: fit a new Version 2 gasket and close the joint per 5.1. With cracks: replace the casing cover (Version 2) and gasket. Do not attempt repair by welding, metal filler or liquid sealant.
6. Report the result of every inspection to Halden with the claim form, including inspections without findings. Halden uses these reports to confirm the affected population.

WARNING / WARNUNG

A cracked gasket seat can release hot liquid under pressure (up to 16 bar, 90 C). Do not approach a leaking casing joint on a running pump to re-tighten bolts. Stop and isolate the pump first.

6 Parts, labour allowance and warranty / Teile, Arbeitszeit und Gewährleistung

Version 2 casing covers and Version 2 casing gaskets are assigned by pump serial number in the Halden partner portal. Order through your Halden service partner, quoting the pump serial number and "SB-2025-009". Mechanical seal kits are not affected by this bulletin; use the kit listed in IOM section 10.

Operation	Labour allowance	Claimable
Inspection 5.2, steps 1 to 3 (joint not opened)	0.5 h	Yes
Re-torque to 38 N-m, 5.2 step 4, incl. drain and refill	1.5 h	Yes
Open joint, dye penetrant test, new gasket, close (5.2 step 5, no cracks)	3.0 h	Yes
Replace casing cover Version 2 and gasket (5.2 step 5, cracks)	4.5 h	Yes
Travel, access, scaffolding, liquid disposal	-	No

Warranty claim code: HW-25-009. Claims are accepted within the pump warranty period (IOM 1.4). As a goodwill measure, Halden will also accept claims outside the warranty period until 2026-12-31 for pumps where tightening to 45 N-m per IOM Rev D is documented. Removed covers with cracks must be held for 90 days and returned on request.

7 Documentation affected / Betroffene Dokumentation

For pumps from serial HCP350-240001, and for pumps fitted with a Version 2 cover, this bulletin supersedes the casing bolt torque given in HDN-IOM-350 Rev D, sections 4.3 (Technical data), 8.4 (Replacing the mechanical seal) and 8.7 (Tightening torques). For earlier serials with the original cover the IOM values remain valid. The correction will be included in IOM Rev E.

Until Rev E is published, keep a copy of this bulletin with the manual at the place of installation and mark printed copies of section 8.7 by hand: "From HCP350-240001: see SB-2025-009".

8 Distribution / Verteiler

- Halden authorised service partners, North America and Europe (acknowledgement in the partner portal within 30 days)
- Halden Sales, regional offices Aalen, Lyon, Katowice and Charlotte NC
- Halden Quality Management (QM-Feld)
- Halden Product Engineering, Centrifugal Pumps
- Halden Spare Parts Service

9 Revision record / Änderungsstand

Rev.	Date	Change	Prepared	Approved
0	2025-06-18	First issue	J. Albrecht	M. Kessler

10 Note on other models / Hinweis zu anderen Baureihen

HCP-200 pumps are not affected. HCP-500 units from serial HCP500-250001 also use the revised casing; torque values for these units are given in torque chart HDN-TC-2025.

Erstellt / Prepared

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Freigegeben / Approved

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Released electronically in Halden PLM. Valid without handwritten signature.

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